PANJAB UNIVERSITY, CHANDIGARH

FACULTY OF SCICNE

SYLLABI

FOR

GEMMOLOGY AND JEWELLERY

ADD-ON COURSE

CERTIFICATE, DIPLOMA AND ADVANCED DIPLOMA

B.A./B.Sc./B.Com.

Examinations,2011

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GEMMOLOGY AND JEWELLERY

OUTLINES OF TESTS SYLLABI AND COURSES OF READING FOR AN ADD-ON CERTIFICATE COURSE IN GEMMOLOGY AND JEWELLERY DURING B.A./B.Sc./B.Com. FOR THE EXAMINATIONS OF 2011.

Outlines of Tests

Paper Code	Paper	Paper Name	Lectures and Practicals/week	Univ. Exam. Marks
	Theory			
101	A	Gemmology	4	75
102	B	Jewellery	4	75
	Practical			
		Based on Paper A & B	4	50

Objective:

The one year Certificate Course in " Gemmology and Jewellery" is an introductory course for the undergraduate students of Colleges consisting of two theory papers and one practical. The objective of the course is to give introduction about the Gems and Jewellery.

PAPER- A

Title : Gemmology **Max. Marks: 75**

Paper Code :101 **Time: 3 Hours**

Course Duration: 90 Lectures

Note for the paper setter: Total of 8 questions may be set covering the whole syllabus Candidates will be required to attempt five questions in all.

1. Definiition of Gem; Introduction to geological occurrences, types of rocks and their mode of occurrence, host, rocks for gen stones.
2. Introduction to Gem minerals and their occurrences in Inida.
3. Physical properties of Gems (Hardness, cleavage, f racture, specific Gravity) Chemical composition and impurities.
4. Introduction to instruments such as low power stereoscope and jeweler's refractometer.

Books:

A Collector's guide to Minerals and Gemstones by John Senbanbas; published by Thames and Hudson, London, 1976.

Dictionary of Gemmology by P.G. Read; published by Dutter Worth Scientific, London, 1982.

PAPER- B

Title : Jewellery **Max. Marks: 75**

Paper Code : 102 **Time: 3 Hours**

Course Duration: 90 Lectures

Note for the paper setter: Total of 8 questions may be set covering the whole syllabus Candidates will be required to attempt five questions in all.

1. Definiition of Jewellery Significance, functions and types of Jewellery.
2. An outlines of the History of Ancient Indian Jerwellery.
3. An Introduction to the methods and Meterial used in the making of Jewellery. (Knewledge of Hand tools used in the making of Jewellery).
4. Jewellery Designing: Drawing patterns, Geometric and Floral, Use of gems in the ornamentation.

Books:

Costume Jewellery – Judith Miller

Indian Jewellery – M.L. Nigam

Traditions Art Jewellery of India.

Eternal India: A Jewelled splendor

Rajasthan's silver Jewellery (Eternal India)

Prtactical (Based on theory Paper A & B)

Max. Marks: 50 Marks

1. Gemmology : Study of Physical and optical Properties of some Precious and Semi Precious stones such as ruby, topaz, garnet and amethyst, Emerald, Zircon.
2. Drawing Skills: Free hand drawing of objects and Figures; Decorative forms and shapes (Geometric & Floral)
3. Aesthetic study of Colour Scheme – Water Colour and Pastel Colours.
4. Designing Jewelllery with Beads, Threads and Clay.
5. Visits to Jewelllery workshops.

OUTLINES OF TESTS SYLLABI AND COURSES OF READING FOR AN ADD-ON DIPLOMA COURSE IN GEMMOLOGY AND JEWELLERY DURING B.A./B.Sc./B.Com. FOR THE EXAMINATIONS OF 2011.

Outlines of Tests

Paper Code	Paper	Paper Name	Lectures and Practicals/week	Univ. Exam. Marks
	Theory			
201	A	Gemmology	4	75
202	B	Jewellery	4	75
	Practical			
		Based on Paper A & B	4	50

Objective:

The one year Diploma Course in "Gemmology and Jewellery" is an introductory course for the undergraduate students of Colleges consisting of two theory papers and one practical. The objective of the course is to give introduction about the Gems and Jewellery.

PAPER- A

Title : Gemmology **Max. Marks: 75**

Paper Code : 201 **Time: 3 Hours**

Course Duration: 90 Lectures

Note for the paper setter: Total of 8 questions may be set covering the whole syllabus Candidates will be required to attempt five questions in all.

1. Gemminerals and their occurrences in India – Native elements, oxides, halides, phosphates and silicates.
2. Geological and geographical occurrences of ores of precious (Au, Pt, Ag) and semi-precious metals in India.
3. Physical and Optical properties of Gems- hardness, weight, cleavage, fracture, specific. Gravity, Fluorescence, Refractive indix, Dispersion, Pleochroism, Cutting and Polishing.
4. Introduction to instruments such as Petrological microscope, polariscope, Dichroscope, colour filters.

Books:

A Collector's guide to Minerals and Gemstones by John Senbanbas; published by Thames and Hudson, London, 1976.

Dictionary of Gemmology by P.G. Read; published by Dutter Worth Scientific, London, 1982.

PAPER- B

Title : Jewellery **Max. Marks: 75**

Paper Code : 202 **Time: 3 Hours**

Course Duration: 90 Lectures

Note for the paper setter: Total of 8 questions may be set covering the whole syllabus Candidates will be required to attempt five questions in all.

1. An outline of the history of Gipta and Medieval India Jewellery.
2. Study of Aesthetic & Valuation of Jewellery products.
3. Study of the physical properties of the precious metals (gold, silver and platinum) and semi-precious metals and alloys (copper & bronze).
4. Various techniques of designing and execution of jewellery items.

Books:

Costume Jewellery – Judith Miller

Indian Jewellery – M.L. Nigam

Traditions Art Jewellery of India.

Eternal India: A Jewelled splendor

Rajasthan's silver Jewellery (Eternal India)

Prtactical (Based on theory Paper A & B)

Max. Marks: 50 Marks

1. Study of Physical properties.
2. Study of Optical properties.
3. Designing Jewelllery with semi-precious stones, metals and alloys.
4. Use of certain tools such as hammer, chisel and files etc. in shaping geometrical and floral patterns employing repouses technique in copper sheet.

OUTLINES OF TESTS SYLLABI AND COURSES OF READING FOR AN ADD-ON ADVANCED DIPLOMA COURSE IN GEMMOLOGY AND JEWELLERY DURING B.A./B.Sc./B.Com. FOR THE EXAMINATIONS OF 2011

Outlines of Tests

Paper Code	Paper	Paper Name	Lectures and Practicalsweek	Univ. Exam. Marks
	Theory			
301	A	Gemmology	4	75
302	B	Jewellery	4	75
	Practical			
		Based on Paper A & B	4	50

Objective:

The one year Diploma Course in " Gemmology and Jewellery" is an introductory course for the undergraduate students of Colleges consisting of two theory papers and one practical. The objective of the course is to give introduction about the Gems and Jewellery.

PAPER- A

Title : Gemmology **Max. Marks: 75**

Paper Code : 301 **Time: 3 Hours**

Course Duration: 90 Lectures

Note for the paper setter: Total of 8 questions may be set covering the whole syllabus Candidates will be required to attempt five questions in all.

1. Prerequisites of a gem mineral: Splendor, durability, rarity, demand, perfection, size , fashioning & portability.
2. Optical basis basis of beauty in gems: colour, selective absorption, dispersion and Interference, luster, transparency & Phenomena.
3. Gem Materials and problems of Identificaition. Natural gem minerals, non-mineral natural gem materials, reconstructed gem, synthetics, assembled or composite stones, cultured pearls, altered natural gemstones.
4. World resources of diamonds; cutting and polishing of diamonds. International trade and restriction. Indian resources and status of polishing and cutting industry with respect to international trade.
5. Status of precious and semi-precious gemstones in Indian national mineral policy.

Books:

Gribble, C.D. , 1988. Rutley's Elements of Mineralogy. CBS Pub., New Delhi.

Liddicoat, R.T., 1969. Handbook of Gem Indentification. Gemological Institute of America.

Sinha, R.K. & Sharma, N.L., 1980. Mineral Economics.. Oxford and IBH Publishing C.

Webster, R., Gems: their sources, description and identification. Butterworths, London.

PRACTICAL

Marks: 25

Identificationh of precious and semi-precious gem stones using various instruments for physical and optical properties.

Exercises related to distribution of gemstones on world map.

PAPER- B

Title : Jewellery **Max. Marks: 75**

Paper Code: 302 **Time: 3 Hours**

Course Duration: 90 Lectures

Note for the paper setter: Total of 8 questions may be set covering the whole syllabus Candidates will be required to attempt five questions in all.

1. An outline of the history of Mughal Jewellery with special reference to the times of Akbar, Jahangir and Shahjahan.

2. Provincial Jewellery – Rajasthani, Gujarati and Punjabi.
3. Tupes of jewellery with special reference to Functional Jewellery and Costume Jewellery.
4. New developments in Jewellery in India after independence with special reference to Designing New Materials, Technology and Functions.

4. Jewellery Designing – Experimental and unconventional designing.

Books:

Verma , S.P.I., Material Culture as Discerned from Mughal Painting, PIHC, Calicut, 1976.

Miller,Judith, Costume Jewellery.

Nigam, M.L., Indian Jewellery.

Majimdar, R.C. (ed.) History and Culture of Indian people, Bhartiya Vidya Bhawan Publication Encyclopedia of World Art.

PRACTICAL

Marks: 25

Designing Jewellery with precious metals and also with unconventional materials such as papier mache, plastic, rubber, bamboo, wood and fibre-glass.

Techniques of care, preservation and conservation of Jewellery items.

Visits to Jewellery making workshops.

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